

CAMBRIDGE INTERNATIONAL AS & A LEVEL CHEMISTRY (9701)

Topic 13 — Introduction to AS Organic Chemistry (Paper 1 Multiple Choice)

Paper 1 (Multiple Choice) Topical Past Paper Pack · 110 Questions & Detailed Solutions

CANDIDATE	Urwah
SYLLABUS	Cambridge International AS Level Chemistry (9701)
COMPONENT	Paper 1: Multiple Choice (1 Hour 15 Minutes Exam Standard)
TOPIC COVERAGE	13.1 Formulas, IUPAC Nomenclature & Functional Groups · 13.2 Organic Reaction Terminology & Mechanisms · 13.3 Shapes of Molecules, Hybridisation & Bonding · 13.4 Isomerism: Structural & Stereoisomerism
TOTAL VOLUME	110 Authentic Multiple Choice Questions (110 Marks · 1 Mark Each)

HOW TO USE THIS PAPER 1 MCQ PACK

1. Timing & Strategy: Spend approximately 1 minute to 1 minute 15 seconds per question under timed test conditions.
2. Answering: For each question, select the one correct option from A, B, C, or D. Apply the elimination method to rule out obvious distractors.
3. Authentic Exam Codes: Every question features authentic Cambridge reference codes (e.g. 9701/12/M/J/23) for cross-study verification.
4. Quick-Check Answer Grid: Use the 110-question matrix immediately following the questions for rapid scoring in under 60 seconds.
5. Comprehensive Explanations: Review the detailed examiner rationale and distractor analysis at the back to eliminate conceptual errors.

SYLLABUS SUBTOPICS COVERED IN THIS PACK

- 13.1 Formulas, Nomenclature & Functional Groups** – Types of chemical formulae (empirical, molecular, general, structural, displayed, skeletal); systematic IUPAC nomenclature
- 13.2 Reaction Terminology & Mechanisms** – Homolytic vs heterolytic fission, free radicals, electrophiles, nucleophiles; classification of reaction types: addition, elimination, substitution, redox
- 13.3 Shapes of Molecules, Hybridisation & Bonding** – Orbital hybridisation (sp^3 tetrahedral 109.5° , sp^2 trigonal planar 120° , sp linear 180°); end-on sigma bond overlap
- 13.4 Isomerism: Structural & Stereoisomerism** – Chain, positional, and functional group isomerism; stereoisomerism: cis-trans / E-Z isomerism in alkenes (restricted rotation); optical isomerism in chiral molecules
- High-Frequency Core Repeats** – The 10 most frequently examined Cambridge Paper 1 questions on Introductory Organic Chemistry (Q101 – Q110).

SUBTOPIC 13.1 — FORMULAS, NOMENCLATURE & FUNCTIONAL GROUPS (Q1 – Q30)

1 [EASY] Types of Chemical Formula — Empirical vs Molecular — 9701/11/M/J/23/Q13

[1]

A volatile liquid compound contains 54.5% carbon, 9.1% hydrogen, and 36.4% oxygen by mass. Its relative molecular mass, M_r , is 88.0.

What are the empirical formula and molecular formula of this compound?

- [A] Empirical formula: CH_2O ; Molecular formula: $\text{C}_3\text{H}_6\text{O}_3$
- [B] Empirical formula: $\text{C}_2\text{H}_4\text{O}$; Molecular formula: $\text{C}_4\text{H}_8\text{O}_2$
- [C] Empirical formula: $\text{C}_4\text{H}_8\text{O}_2$; Molecular formula: $\text{C}_4\text{H}_8\text{O}_2$
- [D] Empirical formula: $\text{C}_2\text{H}_4\text{O}$; Molecular formula: $\text{C}_2\text{H}_4\text{O}$

2 [EASY] Interpreting Skeletal Formulas — 9701/12/M/J/23/Q13

[1]

In a skeletal formula of an organic molecule, what do the vertices (corners) and line ends represent?

- [A] A methyl group ($-\text{CH}_3$) only at every vertex and line end.
- [B] Any heteroatom (oxygen, nitrogen, halogen) unless labelled.
- [C] An unsaturated double bond.
- [D] A carbon atom with the appropriate number of hydrogen atoms attached to satisfy carbon's four-covalent valency.

3 [HARD] IUPAC Nomenclature of Branched Halogenoalkanes — 9701/13/M/J/23/Q13

[1]

What is the correct systematic IUPAC name for the compound $\text{CH}_3\text{-CH}(\text{CH}_3)\text{-CH}(\text{Br})\text{-CH}_2\text{-CH}_3$?

- [A] 3-bromo-2-methylpentane
- [B] 3-bromo-4-methylpentane
- [C] 2-methyl-3-bromopentane
- [D] 1-bromo-1-ethyl-2-methylpropane

4 [EASY] General Formula of Homologous Series — 9701/11/O/N/23/Q13

[1]

Which homologous series has the general formula $\text{C}_n\text{H}_{2n}\text{O}$ and does NOT decolorize aqueous bromine?

- [A] Alkenols (unsaturated alcohols)
- [B] Cycloalkanes
- [C] Non-cyclic aliphatic aldehydes and ketones
- [D] Carboxylic acids

5 [MEDIUM] Identifying Functional Groups in Complex Molecules — 9701/12/O/N/23/Q13

[1]

The molecule phenylephrine has the structure $\text{HO-C}_6\text{H}_4\text{-CH}(\text{OH})\text{-CH}_2\text{-NH-CH}_3$.

Which functional groups are present in phenylephrine?

- [A] Phenol, secondary alcohol, and secondary amine
- [B] Primary alcohol, tertiary amine, and ketone
- [C] Phenol, primary alcohol, and primary amine
- [D] Ester, secondary alcohol, and amide

6 [HARD] Homologous Series Characteristics — 9701/11/M/J/22/Q6

[1]

Which property is TRUE for all members of a homologous series?

- [A] They possess identical physical boiling points.
- [B] They have identical empirical formulas.
- [C] They exhibit identical molecular masses.
- [D] They share the same general formula and identical functional group, displaying a gradation in physical properties.

7 [EASY] Displayed Formula Definition — 9701/12/M/J/222/Q7

[1]

What is explicitly shown in a fully displayed formula of an organic molecule?

- [A] Only the functional group bonds.
- [B] Every single atom and every covalent bond connecting them.
- [C] Only the carbon backbone bonds.
- [D] The 3D spatial orientation of orbitals.

8 [HARD] IUPAC Naming of Alkenes — 9701/13/M/J/223/Q8

[1]

What is the systematic IUPAC name for $(\text{CH}_3)_2\text{C}=\text{CH}-\text{CH}_3$?

- [A] 3-methylbut-2-ene
- [B] 2-methylbut-3-ene
- [C] 2-methylbut-2-ene
- [D] Pent-2-ene

9 [EASY] Functional Group of Esters — 9701/11/M/J/224/Q9

[1]

Which grouping of atoms represents the ester functional group?

- [A] $-\text{CONH}_2$
- [B] $-\text{COO}-$ (or $-\text{COOCH}_2-$)
- [C] $-\text{COCl}$
- [D] $-\text{O}-$

10 [HARD] Molecular Formula Determination — 9701/12/M/J/220/Q10

[1]

A hydrocarbon contains 85.7% carbon by mass and has $M_r = 56$. What is its molecular formula?

- [A] C_4H_8
- [B] C_3H_6
- [C] C_4H_{10}
- [D] C_5H_{10}

11 [EASY] IUPAC Name for Dicarboxylic Acid — 9701/13/M/J/221/Q11

[1]

What is the systematic IUPAC name for $\text{HOOC}-\text{CH}_2-\text{CH}_2-\text{COOH}$?

- [A] Succinic acid
- [B] 1,4-dicarboxyethane
- [C] Butanedioic ester
- [D] Butanedioic acid

12 [HARD] Structural Formula Convention — 9701/11/M/J/222/Q12

[1]

What is the correct condensed structural formula for 2,2-dimethylpropane?

- [A] $\text{CH}_3\text{CH}_2\text{CH}(\text{CH}_3)_2$
- [B] $\text{CH}_3(\text{CH}_2)_3\text{CH}_3$
- [C] $\text{C}(\text{CH}_3)_4$
- [D] $(\text{CH}_3)_2\text{CHCH}_2\text{CH}_3$

13 [EASY] Functional Group of Nitriles — 9701/12/M/J/223/Q13

[1]

Which functional group is present in ethanenitrile?

- [A] $-\text{C}\equiv\text{N}$ (cyano group)
- [B] $-\text{NH}_2$ (amino group)
- [C] $-\text{NO}_2$ (nitro group)
- [D] $-\text{CONH}_2$ (amide group)

14 [HARD] IUPAC Naming of Haloalkanes — 9701/13/M/J/224/Q14

[1]

What is the IUPAC name for $\text{CH}_3\text{-CH}(\text{Cl})\text{-CH}(\text{Cl})\text{-CH}_3$?

- [A] 1,2-dichlorobutane
- [B] 2,2-dichlorobutane
- [C] 2,3-dichlorobutane
- [D] 1,4-dichlorobutane

15 [EASY] Degree of Unsaturation — 9701/11/M/J/220/Q15

[1]

How many degrees of unsaturation (double bond equivalents) are present in $\text{C}_6\text{H}_{10}\text{O}$?

- [A] 1
- [B] 2
- [C] 3
- [D] 0

16 [HARD] Identification of Acyl Chlorides — 9701/12/M/J/221/Q16

[1]

Which formula represents an acyl chloride?

- [A] $\text{CH}_3\text{CH}_2\text{Cl}$
- [B] CH_3CHCl_2
- [C] $\text{CH}_3\text{OCH}_2\text{Cl}$
- [D] CH_3COCl

17 [EASY] IUPAC Name for Branched Alcohol — 9701/13/M/J/222/Q17

[1]

What is the IUPAC name for $\text{CH}_3\text{-C}(\text{CH}_3)(\text{OH})\text{-CH}_2\text{-CH}_3$?

- [A] 3-methylbutan-3-ol
- [B] 2-methylbutan-3-ol
- [C] 2-methylbutan-2-ol
- [D] Pentan-2-ol

18 [HARD] Classification of Alcohols — 9701/11/M/J/223/Q18

[1]

Which molecule is a tertiary alcohol?

- [A] 2-methylpropan-2-ol
- [B] Propan-2-ol
- [C] 2-methylpropan-1-ol
- [D] Butan-1-ol

19 [EASY] Functional Groups in Amino Acids — 9701/12/M/J/224/Q19

[1]

Which two functional groups are present in all 2-amino acids?

- [A] Amide and alcohol
- [B] Ester and amine
- [C] Aldehyde and ketone
- [D] Carboxylic acid ($-\text{COOH}$) and primary amine ($-\text{NH}_2$)

20 [HARD] General Formula of Alkanes — 9701/13/M/J/220/Q20

[1]

What is the general formula for all open-chain saturated hydrocarbons (alkanes)?

- [A] C_nH_{2n}
- [B] $\text{C}_n\text{H}_{2n+2}$
- [C] $\text{C}_n\text{H}_{2n-2}$
- [D] $\text{C}_n\text{H}_{2n+1}$

21 [EASY] General Formula of Cycloalkanes — 9701/11/M/J/221/Q21

[1]

What is the general formula for monocyclic cycloalkanes?

- [A] C_nH_{2n+2}
- [B] C_nH_{2n}
- [C] C_nH_{2n-2}
- [D] C_nH_{2n-4}

22 [HARD] Naming of Cyclic Hydrocarbons — 9701/12/M/J/222/Q22

[1]

What is the IUPAC name of a saturated 6-membered ring hydrocarbon?

- [A] Benzene
- [B] Hexene
- [C] Cyclohexene
- [D] Cyclohexane

23 [EASY] Identification of Ketones — 9701/13/M/J/223/Q23

[1]

Which condensed formula represents a ketone?

- [A] CH_3COCH_3
- [B] CH_3CH_2CHO
- [C] CH_3COOCH_3
- [D] CH_3OCH_3

24 [HARD] Identification of Ethers — 9701/11/M/J/224/Q24

[1]

What is the functional group present in $CH_3-O-CH_2CH_3$?

- [A] Ester
- [B] Alcohol
- [C] Ether
- [D] Peroxide

25 [EASY] IUPAC Name of Aldehyde — 9701/12/M/J/220/Q25

[1]

What is the IUPAC name for $CH_3-CH(CH_3)-CH_2-CHO$?

- [A] 3-methylbutanal
- [B] 2-methylbutanal
- [C] 3-methylbutanone
- [D] Pentanal

26 [HARD] Skeletal Formula of Benzene — 9701/13/M/J/221/Q26

[1]

How is benzene represented in skeletal notation?

- [A] A pentagon with a double bond
- [B] A zig-zag line of 6 carbons
- [C] An open ring of 6 CH_2 groups
- [D] A regular hexagon with an inscribed circle (or alternating double bonds)

27 [EASY] Empirical Formula from Combustion Data — 9701/11/M/J/222/Q27

[1]

Complete combustion of 0.20 mol of a hydrocarbon produces 0.80 mol CO_2 and 0.80 mol H_2O . What is the hydrocarbon?

- [A] C_4H_{10}
- [B] C_4H_8
- [C] C_2H_4
- [D] C_3H_6

28 [HARD] Carboxylic Acid Salt Naming — 9701/12/M/J/223/Q28

[1]

What is the IUPAC name for $\text{CH}_3\text{CH}_2\text{COONa}$?

- [A] Sodium propionate
- [B] Sodium ethanoate
- [C] Sodium propanoate
- [D] Sodium butanoate

29 [EASY] Primary Amine Nomenclature — 9701/13/M/J/224/Q29

[1]

What is the systematic IUPAC name for $\text{CH}_3\text{-CH}_2\text{-CH}_2\text{-NH}_2$?

- [A] Propylamine
- [B] Propan-1-amine
- [C] Ethylmethanamine
- [D] 1-aminopropane

30 [HARD] Hybrid Functional Molecules — 9701/11/M/J/220/Q30

[1]

Which functional groups are present in $\text{CH}_2=\text{CH-COOH}$?

- [A] Alkene ($\text{C}=\text{C}$) and carboxylic acid ($-\text{COOH}$)
- [B] Alkyne and ester
- [C] Alcohol and aldehyde
- [D] Alkane and ketone

SUBTOPIC 13.2 — REACTION TERMINOLOGY & MECHANISMS (Q31 – Q55)**31** [EASY] Homolytic vs Heterolytic Bond Fission — 9701/11/M/J/23/Q14

[1]

What occurs during homolytic fission of a covalent bond?

- [A] One atom takes both shared electrons, forming a cation and an anion.
- [B] A pair of electrons is transferred to an electrophile to form a dative bond.
- [C] The bond vibrates until it dissociates into two ions of opposite charge.
- [D] The covalent bond breaks symmetrically such that each departing atom retains one of the two shared bonding electrons, producing two uncharged free radicals.

32 [EASY] Definition and Behavior of an Electrophile — 9701/12/M/J/23/Q14

[1]

Which statement correctly defines an electrophile in organic reaction mechanisms?

- [A] A species with a lone pair of electrons that donates them to form a dative covalent bond.
- [B] A neutral species with an unpaired electron that abstracts a hydrogen atom.
- [C] An electron-deficient species (either positively charged or possessing an electron-deficient dipole δ^+) that attacks regions of high electron density by accepting an electron pair.
- [D] A proton donor according to the Bronsted-Lowry acid-base definition.

33 [EASY] Definition and Behavior of a Nucleophile — 9701/13/M/J/23/Q14

[1]

Which species can act as a nucleophile in organic substitution reactions?

- [A] $:\text{OH}^-$, $:\text{CN}^-$, $:\text{NH}_3$, and H_2O : (species possessing a non-bonding lone pair of electrons)
- [B] Br^+ , NO_2^+ , and AlCl_3 (electron-deficient Lewis acids)
- [C] $\text{Cl}^\cdot$ and $\text{CH}_3^\cdot$ (free radicals with single unpaired electrons)
- [D] CH_4 and C_2H_6 (saturated alkanes with no lone pairs)

34 [MEDIUM] Classification of Reaction Types — 9701/11/O/N/23/Q14

[1]

The conversion of bromoethane into ethene by heating with ethanolic potassium hydroxide is classified as which type of reaction?

- [A] Electrophilic addition
- [B] Nucleophilic substitution
- [C] Elimination
- [D] Free radical substitution

35 [HARD] Free Radical Substitution Mechanism Stages — 9701/12/O/N/23/Q14

[1]

In the photochemical chlorination of methane, which step represents a propagation reaction?

- [A] $\text{Cl}_2 \rightarrow 2\text{Cl}\cdot$ (under UV light)
- [B] $\text{CH}_3\cdot + \text{Cl}_2 \rightarrow \text{CH}_3\text{Cl} + \text{Cl}\cdot$
- [C] $\text{CH}_3\cdot + \text{Cl}\cdot \rightarrow \text{CH}_3\text{Cl}$
- [D] $\text{CH}_3\cdot + \text{CH}_3\cdot \rightarrow \text{C}_2\text{H}_6$

36 [HARD] Curly Arrow Convention — 9701/11/M/J/221/Q36

[1]

In organic reaction mechanisms, what does a standard double-barbed curly arrow represent?

- [A] The physical movement of an atomic nucleus.
- [B] The shift of a single unpaired electron.
- [C] The direction of thermodynamic heat flow.
- [D] The movement of a pair of electrons from a lone pair or covalent bond to an atom or bond.

37 [EASY] Fish-hook Arrow Convention — 9701/12/M/J/222/Q37

[1]

What does a single-barbed (fish-hook) curly arrow denote in organic chemistry?

- [A] The transfer of two electrons in an elimination reaction.
- [B] A resonance contributor.
- [C] The movement of a single unpaired electron.
- [D] A coordinate bond.

38 [HARD] Carbocation Stability Order — 9701/13/M/J/223/Q38

[1]

What is the correct order of carbocation stability from most stable to least stable?

- [A] Tertiary (3°) > Secondary (2°) > Primary (1°) > Methyl
- [B] Methyl > Primary (1°) > Secondary (2°) > Tertiary (3°)
- [C] Primary > Tertiary > Secondary > Methyl
- [D] Secondary > Tertiary > Primary > Methyl

39 [EASY] Electrophilic Addition Mechanism — 9701/11/M/J/224/Q39

[1]

Which reaction proceeds via an electrophilic addition mechanism?

- [A] Methane reacting with chlorine in UV light.
- [B] Bromoethane reacting with aqueous sodium hydroxide.
- [C] Ethanol reacting with acidified potassium dichromate.
- [D] Ethene reacting with hydrogen bromide to form bromoethane.

40 [HARD] Nucleophilic Substitution Mechanism — 9701/12/M/J/220/Q40

[1]

Which reaction proceeds via a nucleophilic substitution mechanism?

- [A] Propene reacting with steam over H_3PO_4 .
- [B] 1-bromopropane reacting with aqueous NaOH to form propan-1-ol.
- [C] Ethanol reacting with hot concentrated sulfuric acid.
- [D] Ethane reacting with bromine under UV light.

41 [EASY] Hydrolysis Reaction Definition — 9701/13/M/J/221/Q41**[1]**

What defines a hydrolysis reaction in organic chemistry?

- [A] The simple physical dissolution of a substance in water.
- [B] A reaction where a covalent bond in a molecule is cleaved by the chemical action of water (often catalyzed by acid or base).
- [C] The removal of water from a molecule to form an alkene.
- [D] The hydration of an alkene across a double bond.

42 [HARD] Condensation Reaction Definition — 9701/11/M/J/222/Q42**[1]**

What characterizes a condensation reaction?

- [A] A gas turning into a liquid upon cooling.
- [B] Addition of hydrogen across a multiple bond.
- [C] The breakdown of a polymer into monomers.
- [D] Two molecules combine to form a larger single molecule with the elimination of a small molecule such as H₂O or HCl.

43 [EASY] Reduction of Carbonyl Compounds — 9701/12/M/J/223/Q43**[1]**

Which reagent reduces an aldehyde or ketone to an alcohol?

- [A] Sodium borohydride, NaBH₄ (in aqueous ethanol)
- [B] Acidified potassium dichromate(VI)
- [C] Concentrated sulfuric acid
- [D] Tollens' reagent

44 [HARD] Oxidation in Organic Chemistry — 9701/13/M/J/224/Q44**[1]**

In organic chemistry, what does oxidation typically involve?

- [A] The addition of hydrogen across a C=C bond.
- [B] The loss of an electron pair by a nucleophile.
- [C] An increase in the number of carbon-oxygen bonds and/or a decrease in carbon-hydrogen bonds.
- [D] The substitution of chlorine by fluorine.

45 [EASY] Inductive Effect Definition — 9701/11/M/J/220/Q45**[1]**

What is the positive inductive effect (+I effect)?

- [A] The electron-donating ability of alkyl groups through sigma bonds toward a partially positive carbon center.
- [B] The withdrawal of electrons by highly electronegative atoms like fluorine.
- [C] The delocalization of pi electrons in aromatic rings.
- [D] The polarization of light by chiral centers.

46 [HARD] Markovnikov's Rule Basis — 9701/12/M/J/221/Q46**[1]**

What is the underlying mechanistic basis for Markovnikov's rule in alkene addition?

- [A] The halogen atom always prefers to bond to the carbon with more hydrogen atoms.
- [B] Steric hindrance prevents addition at tertiary carbons.
- [C] Hydrogen bonds direct the incoming electrophile.
- [D] The reaction proceeds predominantly via the more stable carbocation intermediate.

47 [EASY] S_N1 Mechanism Characteristics — 9701/13/M/J/222/Q47**[1]**

Which feature is characteristic of an S_N1 nucleophilic substitution reaction?

- [A] A concerted one-step mechanism with a five-coordinate transition state.
- [B] A two-step mechanism involving a planar carbocation intermediate, typical for tertiary halogenoalkanes.
- [C] Second-order kinetics dependent on both substrate and nucleophile concentrations.
- [D] Complete inversion of optical configuration.

48 [HARD] SN2 Mechanism Characteristics — 9701/11/M/J/223/Q48

[1]

Which feature is characteristic of an SN2 nucleophilic substitution reaction?

- [A] Formation of a stable carbocation intermediate.
- [B] First-order kinetics independent of nucleophile concentration.
- [C] A one-step concerted mechanism with backside attack causing Walden inversion of configuration.
- [D] Racemisation of chiral optical centers.

49 [EASY] Reflux Technique in Organic Synthesis — 9701/12/M/J/224/Q49

[1]

Why is a reaction mixture heated under reflux during organic preparations?

- [A] To increase the atmospheric pressure above the liquid.
- [B] To heat the mixture at its boiling temperature without losing volatile reactants or products as vapor.
- [C] To separate two liquids with different boiling points.
- [D] To freeze the reaction intermediate.

50 [HARD] Role of UV Light in Halogenation — 9701/13/M/J/220/Q50

[1]

What is the role of ultraviolet (UV) radiation in the reaction between alkanes and halogens?

- [A] It provides the energy required to homolytically cleave the X-X bond to produce halogen free radicals.
- [B] It ionizes the alkane into a carbocation.
- [C] It acts as a heterogeneous catalyst.
- [D] It polarizes the carbon-hydrogen bond.

51 [EASY] Distinguishing Addition from Substitution — 9701/11/M/J/221/Q51

[1]

How does an addition reaction fundamentally differ from a substitution reaction?

- [A] Addition only occurs in alkanes.
- [B] Substitution always breaks double bonds.
- [C] Addition requires high temperatures whereas substitution occurs at 0 °C.
- [D] In addition, two reactant molecules combine to form a single product with 100% atom economy; in substitution, an atom or group is replaced with formation of a byproduct.

52 [HARD] Free Radical Termination Step — 9701/12/M/J/222/Q52

[1]

Which reaction is an example of a termination step in a free radical mechanism?

- [A] $\text{Cl} \cdot + \text{CH}_4 \rightarrow \cdot\text{CH}_3 + \text{HCl}$
- [B] $\text{Cl}_2 \rightarrow 2\text{Cl} \cdot$
- [C] $2\text{Cl} \cdot \rightarrow \text{Cl}_2$
- [D] $\cdot\text{CH}_3 + \text{Cl}_2 \rightarrow \text{CH}_3\text{Cl} + \text{Cl} \cdot$

53 [EASY] Nucleophilic Attack at Carbonyls — 9701/13/M/J/223/Q53

[1]

Why is the carbonyl carbon in ethanal susceptible to nucleophilic attack?

- [A] Oxygen is more electronegative than carbon, polarizing the C=O bond so the carbon carries a significant partial positive charge (δ^+).
- [B] The carbon has an expanded octet of 10 electrons.
- [C] The carbonyl bond is completely non-polar.
- [D] Carbon has two non-bonding lone pairs.

54 [HARD] Steric Hindrance in S_N2 — 9701/11/M/J/224/Q54

[1]

Why do tertiary halogenoalkanes NOT react via the S_N2 mechanism?

- [A] Tertiary halogenoalkanes cannot form carbocations.
- [B] The C-X bond enthalpy is too high to break.
- [C] Three bulky alkyl groups surround the central carbon, sterically blocking the incoming nucleophile from attacking from the rear.
- [D] Tertiary carbons have no vacant orbitals.

55 [EASY] Elimination vs Substitution Regiochemistry — 9701/12/M/J/220/Q55

[1]

What reaction conditions favor elimination over substitution for 2-bromopropane?

- [A] Shaking with dilute aqueous sodium hydroxide at room temperature.
- [B] Heating with concentrated ethanolic potassium hydroxide at high temperature.
- [C] Treating with cold acidified water.
- [D] Exposing to chlorine in ultraviolet light.

SUBTOPIC 13.3 — SHAPES OF MOLECULES, HYBRIDISATION & BONDING (Q56 – Q75)**56** [EASY] Hybridisation in Ethene — 9701/11/M/J/23/Q15

[1]

What is the orbital hybridisation of the carbon atoms in ethene, C_2H_4 , and what is the approximate H-C-H bond angle?

- [A] sp^3 hybridised; bond angle 109.5°
- [B] sp hybridised; bond angle 180°
- [C] dsp^2 hybridised; bond angle 90°
- [D] sp^2 hybridised; bond angle approximately 120°

57 [HARD] Nature and Overlap of Sigma vs Pi Bonds — 9701/12/M/J/23/Q15

[1]

Which statement accurately contrasts a sigma (σ) bond with a pi (π) bond in organic molecules?

- [A] A pi bond is formed by end-on overlap and is significantly stronger than a sigma bond.
- [B] Sigma bonds can only exist between identical atoms whereas pi bonds only form between different elements.
- [C] A sigma bond is formed by direct end-on (head-to-head) overlap of orbitals with electron density concentrated along the internuclear axis; a pi bond is formed by sideways (lateral) overlap of parallel p orbitals with electron density above and below the axis.
- [D] Free rotation is possible around a pi bond without breaking it.

58 [MEDIUM] Hybridisation and Geometry in Ethyne — 9701/13/M/J/23/Q15

[1]

In the ethyne molecule ($H-C\equiv C-H$), what is the hybridisation of the carbon atoms, the number of sigma and pi bonds, and the molecular geometry?

- [A] sp hybridisation; 3 sigma bonds and 2 pi bonds; linear geometry with 180° bond angles
- [B] sp^2 hybridisation; 2 sigma bonds and 3 pi bonds; planar geometry with 120° bond angles
- [C] sp^3 hybridisation; 5 sigma bonds and 0 pi bonds; tetrahedral geometry with 109.5° bond angles
- [D] sp hybridisation; 1 sigma bond and 4 pi bonds; bent geometry with 104.5° bond angles

59 [EASY] Bond Angles in Propene — 9701/13/M/J/224/Q59

[1]

What are the approximate bond angles around C_1 ($CH_2=$), C_2 ($=CH-$), and C_3 ($-CH_3$) in propene, $CH_2=CH-CH_3$?

- [A] C_1 : 109.5° , C_2 : 120° , C_3 : 180°
- [B] C_1 : 180° , C_2 : 120° , C_3 : 109.5°
- [C] C_1 : 120° , C_2 : 109.5° , C_3 : 120°
- [D] C_1 : 120° , C_2 : 120° , C_3 : 109.5°

60 [HARD] Number of Sigma and Pi Bonds in Propanone — 9701/11/M/J/220/Q60**[1]**How many sigma bonds and pi bonds are present in a molecule of propanone, CH_3COCH_3 ?

- [A] 8 sigma bonds and 2 pi bonds
- [B] 9 sigma bonds and 1 pi bond
- [C] 10 sigma bonds and 0 pi bonds
- [D] 7 sigma bonds and 2 pi bonds

61 [EASY] Planarity of Carbonyl Group — 9701/12/M/J/221/Q61**[1]**Why is the carbonyl group ($>\text{C}=\text{O}$) planar?

- [A] The carbon is sp^3 hybridised with two lone pairs.
- [B] The carbonyl carbon is sp^2 hybridised with three coplanar sigma bonds directed at 120° angles.
- [C] The carbon is sp hybridised with a linear axis.
- [D] Double bonds always cause tetrahedral geometry.

62 [HARD] Restricted Rotation Around Double Bonds — 9701/13/M/J/222/Q62**[1]**

Why is free rotation impossible around the carbon-carbon double bond in alkenes at room temperature?

- [A] The sigma bond is too short to allow rotation.
- [B] The hydrogen atoms collide with each other.
- [C] The carbon atoms carry identical partial charges that repel.
- [D] Rotation would disrupt the parallel sideways overlap of the p orbitals, requiring $\sim 260 \text{ kJ mol}^{-1}$ to break the pi bond.

63 [EASY] Number of Pi Bonds in Benzene — 9701/11/M/J/223/Q63**[1]**How many pi electrons are delocalised in the aromatic ring of benzene, C_6H_6 ?

- [A] 6 pi electrons (one from each sp^2 hybridised carbon atom)
- [B] 12 pi electrons
- [C] 3 pi electrons
- [D] 4 pi electrons

64 [HARD] Bond Length Comparison — 9701/12/M/J/224/Q64**[1]**

Which sequence lists carbon-carbon bonds in order of DECREASING bond length (longest to shortest)?

- [A] $\text{C}\equiv\text{C} > \text{C}=\text{C} > \text{C}-\text{C}$
- [B] $\text{C}=\text{C} > \text{C}-\text{C} > \text{C}\equiv\text{C}$
- [C] $\text{C}-\text{C}$ (single) $> \text{C}=\text{C}$ (double) $> \text{C}\equiv\text{C}$ (triple)
- [D] $\text{C}-\text{C} > \text{C}\equiv\text{C} > \text{C}=\text{C}$

65 [EASY] Bond Energy Comparison — 9701/13/M/J/220/Q65**[1]**

Which sequence ranks carbon-carbon bonds in order of INCREASING bond dissociation enthalpy (weakest to strongest)?

- [A] $\text{C}-\text{C}$ (single) $< \text{C}=\text{C}$ (double) $< \text{C}\equiv\text{C}$ (triple)
- [B] $\text{C}\equiv\text{C} < \text{C}=\text{C} < \text{C}-\text{C}$
- [C] $\text{C}=\text{C} < \text{C}-\text{C} < \text{C}\equiv\text{C}$
- [D] $\text{C}-\text{C} < \text{C}\equiv\text{C} < \text{C}=\text{C}$

66 [HARD] Hybridisation of Nitrile Carbon — 9701/11/M/J/221/Q66**[1]**What is the hybridisation of the carbon and nitrogen atoms in ethanenitrile, $\text{CH}_3-\text{C}\equiv\text{N}$?

- [A] The carbon is sp^2 and nitrogen is sp^3
- [B] The carbon is sp^3 and nitrogen is sp
- [C] Both atoms are sp^2 hybridised
- [D] Both the nitrile carbon and nitrogen are sp hybridised

67 [EASY] Coplanar Atoms in Ethene — 9701/12/M/J/222/Q67

[1]

How many atoms lie in the exact same geometric plane in an ethene molecule (C_2H_4)?

- [A] Only the 2 carbon atoms
- [B] All 6 atoms (both carbons and all four hydrogens)
- [C] 4 atoms (the two carbons and two hydrogens)
- [D] 5 atoms

68 [HARD] Carbon Atom Geometry in Diamond vs Graphite — 9701/13/M/J/223/Q68

[1]

What is the hybridisation of carbon atoms in diamond and graphite respectively?

- [A] Diamond: sp^2 ; Graphite: sp^3
- [B] Diamond: sp ; Graphite: sp^2
- [C] Diamond: sp^3 (tetrahedral); Graphite: sp^2 (trigonal planar sheets)
- [D] Diamond: sp^3 ; Graphite: sp

69 [EASY] Sigma Bonds in Buta-1,3-diene — 9701/11/M/J/224/Q69

[1]

How many sigma bonds are present in a molecule of buta-1,3-diene ($\text{CH}_2=\text{CH}-\text{CH}=\text{CH}_2$)?

- [A] 11 sigma bonds
- [B] 9 sigma bonds
- [C] 7 sigma bonds
- [D] 8 sigma bonds

70 [HARD] Shape Around Oxygen in Alcohols — 9701/12/M/J/220/Q70

[1]

What is the shape and approximate bond angle around the oxygen atom in methanol ($\text{CH}_3-\text{O}-\text{H}$)?

- [A] Bent (V-shaped) with bond angle approximately $104.5^\circ - 109^\circ$
- [B] Trigonal planar with bond angle 120°
- [C] Linear with bond angle 180°
- [D] Tetrahedral with bond angle 109.5°

71 [EASY] Shape Around Nitrogen in Amines — 9701/13/M/J/221/Q71

[1]

What is the molecular geometry around the nitrogen atom in methylamine, CH_3NH_2 ?

- [A] Trigonal planar with bond angle 120°
- [B] Tetrahedral with bond angle 109.5°
- [C] T-shaped with bond angle 90°
- [D] Trigonal pyramidal with bond angle approximately 107°

72 [HARD] Number of Pi Bonds in Carbon Dioxide — 9701/11/M/J/222/Q72

[1]

How many sigma and pi bonds are present in carbon dioxide, $\text{O}=\text{C}=\text{O}$?

- [A] 4 sigma bonds and 0 pi bonds
- [B] 1 sigma bond and 3 pi bonds
- [C] 2 sigma bonds and 2 pi bonds
- [D] 3 sigma bonds and 1 pi bond

73 [EASY] Hybridisation in Carbocations — 9701/12/M/J/223/Q73

[1]

What is the orbital hybridisation and geometry of the positively charged carbon in a methyl carbocation, $+\text{CH}_3$?

- [A] sp^2 hybridised with trigonal planar geometry (120°)
- [B] sp^3 hybridised with pyramidal geometry (107°)
- [C] sp hybridised with linear geometry (180°)
- [D] unhybridised with spherical geometry

74 [HARD] Bond Length in Benzene vs Cyclohexene — 9701/13/M/J/224/Q74**[1]**

How does the carbon-carbon bond length in benzene compare with ethane and ethene?

- [A] It alternates between short 0.134 nm and long 0.154 nm.
- [B] It is shorter than a triple bond (0.110 nm).
- [C] It is intermediate between C-C single (0.154 nm) and C=C double (0.134 nm), with all six C-C bonds equal at 0.139 nm.
- [D] It is identical to the single bond in ethane (0.154 nm).

75 [EASY] Pi Bond Formation Orbitals — 9701/11/M/J/220/Q75**[1]**Which atomic orbitals overlap to form the pi bond in methanal, $\text{H}_2\text{C}=\text{O}$?

- [A] End-on overlap of sp^2 hybrid orbitals.
- [B] Sideways overlap of the unhybridised 2p orbital of carbon with a 2p orbital of oxygen.
- [C] Overlap of a 1s hydrogen orbital with an sp^2 carbon orbital.
- [D] Overlap of two 2s orbitals.

SUBTOPIC 13.4 — STRUCTURAL & STEREOISOMERISM (Q76 – Q100)**76 [EASY]** Classification of Structural Isomerism — 9701/11/M/J/23/Q16**[1]**

What are the three main types of structural isomerism recognized in AS Level Chemistry?

- [A] Cis-trans isomerism, optical isomerism, and conformational isomerism
- [B] Enantiomerism, diastereomerism, and geometric isomerism
- [C] Addition isomerism, substitution isomerism, and elimination isomerism
- [D] Chain isomerism, positional isomerism, and functional group isomerism

77 [EASY] Functional Group Isomerism Pair — 9701/12/M/J/23/Q16**[1]**Which pair of compounds represents functional group isomers with the molecular formula $\text{C}_3\text{H}_6\text{O}$?

- [A] Propan-1-ol and propan-2-ol
- [B] Propanoic acid and methyl ethanoate
- [C] Propanal (aldehyde) and propanone (ketone)
- [D] Cyclopropane and propene

78 [HARD] Conditions for Cis-Trans (Geometric) Isomerism — 9701/13/M/J/23/Q16**[1]**

Which two conditions MUST be satisfied for an alkene to exhibit cis-trans (geometric) stereoisomerism?

- [A] Restricted rotation around the carbon-carbon double bond ($\text{C}=\text{C}$) AND two different groups attached to EACH carbon of the double bond.
- [B] A chiral center with four different groups attached AND a carbon-carbon double bond.
- [C] At least one triple bond AND identical groups on opposite sides of the molecule.
- [D] A saturated carbon chain of at least five carbon atoms.

79 [HARD] Chirality and Optical Activity — 9701/11/O/N/23/Q16**[1]**

What defines a chiral carbon atom (stereocenter) in an organic molecule?

- [A] A carbon atom bonded to a double bond and two hydrogen atoms.
- [B] A carbon atom that forms coordinate bonds with transition metal ions.
- [C] A carbon atom located at the center of an aromatic benzene ring.
- [D] A carbon atom bonded to four completely different atoms or groups of atoms, resulting in non-superimposable mirror images (enantiomers).

80 [MEDIUM] Identifying Chiral Centers in Alcohols — 9701/12/O/N/23/Q16

[1]

Which four-carbon alcohol contains a chiral carbon atom and exists as a pair of optical isomers?

- [A] Butan-1-ol, $\text{CH}_3\text{-CH}_2\text{-CH}_2\text{-CH}_2\text{OH}$
- [B] Butan-2-ol, $\text{CH}_3\text{-CH(OH)-CH}_2\text{-CH}_3$
- [C] 2-methylpropan-1-ol, $(\text{CH}_3)_2\text{CH-CH}_2\text{OH}$
- [D] 2-methylpropan-2-ol, $(\text{CH}_3)_3\text{C-OH}$

81 [EASY] Structural Isomers of Butane — 9701/11/M/J/221/Q81

[1]

How many structural isomers exist for the alkane with molecular formula C_4H_{10} ?

- [A] 3
- [B] 2 (butane and 2-methylpropane)
- [C] 4
- [D] 1

82 [HARD] Structural Isomers of Pentane — 9701/12/M/J/222/Q82

[1]

How many structural isomers exist for the alkane with molecular formula C_5H_{12} ?

- [A] 2
- [B] 4
- [C] 5
- [D] 3 (pentane, 2-methylbutane, and 2,2-dimethylpropane)

83 [EASY] Positional Isomers of Dichlorobenzene — 9701/13/M/J/223/Q83

[1]

How many positional isomers exist for dichlorobenzene ($\text{C}_6\text{H}_4\text{Cl}_2$)?

- [A] 3 (1,2-dichlorobenzene, 1,3-dichlorobenzene, and 1,4-dichlorobenzene)
- [B] 2
- [C] 4
- [D] 1

84 [HARD] Functional Group Isomers of Carboxylic Acids — 9701/11/M/J/224/Q84

[1]

Which functional group class is isomeric with aliphatic carboxylic acids of formula $\text{C}_n\text{H}_{2n}\text{O}_2$?

- [A] Aldehydes
- [B] Alcohols
- [C] Esters
- [D] Ethers

85 [EASY] Cis-Trans in But-2-ene — 9701/12/M/J/220/Q85

[1]

Why does but-2-ene show cis-trans isomerism while but-1-ene does not?

- [A] In but-2-ene, each double-bond carbon has -H and -CH_3 ; in but-1-ene, C_1 has two identical -H atoms.
- [B] But-1-ene has a higher boiling point than but-2-ene.
- [C] But-2-ene contains a chiral carbon atom.
- [D] But-1-ene has sp hybridised carbons.

86 [HARD] Physical Property Differences of Cis-Trans Isomers — 9701/13/M/J/221/Q86

[1]

How do cis- and trans-1,2-dichloroethene differ in physical properties such as boiling point and dipole moment?

- [A] Both isomers are non-polar with identical boiling points.
- [B] The trans isomer is highly polar with a higher boiling point.
- [C] The cis isomer has a lower melting point due to hydrogen bonding.
- [D] The cis isomer is polar with a higher boiling point; the trans isomer is non-polar (dipoles cancel) with a lower boiling point.

87 [EASY] Chirality in Halogenoalkanes — 9701/11/M/J/222/Q87

[1]

Which molecule contains a chiral center?

- [A] 1-bromobutane
- [B] 2-bromobutane
- [C] 2-bromopropane
- [D] 2-bromo-2-methylpropane

88 [HARD] Racemic Mixture Definition — 9701/12/M/J/223/Q88

[1]

What is an equimolar (50:50) mixture of two optical enantiomers called, and what is its net optical rotation?

- [A] A meso compound; it rotates light clockwise.
- [B] A pure enantiomer; it rotates light 180°.
- [C] A racemic mixture (or racemate); it exhibits zero net optical rotation because the equal and opposite rotations cancel.
- [D] A diastereomer; it has a high specific rotation.

89 [EASY] Number of Optical Isomers for 1 Chiral Center — 9701/13/M/J/224/Q89

[1]

An organic molecule has exactly one chiral carbon atom. How many stereoisomers exist for this molecule?

- [A] 4
- [B] 2 (a pair of non-superimposable enantiomers)
- [C] 3
- [D] 1

90 [HARD] Optical Activity Detection — 9701/11/M/J/220/Q90

[1]

How can two optical enantiomers be distinguished experimentally in a laboratory?

- [A] By measuring the direction of rotation of plane-polarized light using a polarimeter.
- [B] By measuring their boiling points.
- [C] By observing their solubility in water.
- [D] By comparing their mass spectra molecular ion peaks.

91 [EASY] Isomerism in Pent-2-ene — 9701/12/M/J/221/Q91

[1]

What type of stereoisomerism is exhibited by pent-2-ene, $\text{CH}_3\text{-CH=CH-CH}_2\text{-CH}_3$?

- [A] Optical isomerism
- [B] Conformational isomerism only
- [C] No isomerism is possible
- [D] Cis-trans (E/Z) isomerism

92 [HARD] Chirality in 3-methylhexane — 9701/13/M/J/222/Q92

[1]

Which carbon atom in 3-methylhexane, $\text{CH}_3\text{-CH}_2\text{-CH(CH}_3\text{)-CH}_2\text{-CH}_2\text{-CH}_3$, is the chiral stereocenter?

- [A] Carbon-2
- [B] Carbon-4
- [C] Carbon-3
- [D] Carbon-1

93 [EASY] Positional Isomers of Propanol — 9701/11/M/J/223/Q93

[1]

What type of isomerism exists between propan-1-ol and propan-2-ol?

- [A] Positional isomerism
- [B] Chain isomerism
- [C] Functional group isomerism
- [D] Stereoisomerism

94 [HARD] Chain Isomerism Example — 9701/12/M/J/224/Q94

[1]

Which pair of molecules represents chain isomers?

- [A] Propan-1-ol and propan-2-ol
- [B] Ethanoic acid and methyl methanoate
- [C] Pentane and 2,2-dimethylpropane
- [D] But-1-ene and but-2-ene

95 [EASY] Total Isomers of $C_3H_6Cl_2$ — 9701/13/M/J/220/Q95

[1]

How many structural isomers exist for the dichloropropane molecule $C_3H_6Cl_2$?

- [A] 3
- [B] 4 (1,1-dichloropropane, 1,2-dichloropropane, 1,3-dichloropropane, and 2,2-dichloropropane)
- [C] 5
- [D] 2

96 [HARD] Functional Group Isomerism of Alkenes and Cycloalkanes — 9701/11/M/J/221/Q96

[1]

What type of isomerism is demonstrated by hex-1-ene and cyclohexane (both C_6H_{12})?

- [A] Chain isomerism
- [B] Positional isomerism
- [C] Optical isomerism
- [D] Functional group isomerism

97 [EASY] Stereoisomerism in 1,2-dimethylcyclopropane — 9701/12/M/J/222/Q97

[1]

Which type of stereoisomerism can be exhibited by 1,2-dimethylcyclopropane?

- [A] Neither cis-trans nor optical isomerism
- [B] Chain isomerism only
- [C] Both cis-trans and optical isomerism
- [D] Functional group isomerism only

98 [HARD] Lack of Chirality in Glycine — 9701/13/M/J/223/Q98

[1]

Why is glycine (2-aminoethanoic acid, H_2N-CH_2-COOH) the only standard amino acid that is optically inactive?

- [A] Its alpha-carbon is bonded to two identical hydrogen atoms, so it is achiral (lacks 4 different groups).
- [B] It contains an amide bond instead of an amine.
- [C] It exists purely as a gaseous free radical.
- [D] It has a double bond between nitrogen and carbon.

99 [EASY] Isomerism of C_2H_6O — 9701/11/M/J/224/Q99

[1]

What are the two structural isomers of C_2H_6O , and what type of isomerism do they exhibit?

- [A] Ethanol and ethanal; positional isomerism
- [B] Ethanol and ethanoic acid; chain isomerism
- [C] Methoxymethane and dimethyl ketone; geometric isomerism
- [D] Ethanol (CH_3CH_2OH) and methoxymethane (CH_3OCH_3); functional group isomerism

100 [HARD] E/Z Nomenclature System Basis — 9701/12/M/J/220/Q100

[1]

What is the Cahn-Ingold-Prelog (CIP) priority rule used in E/Z isomerism based on?

- [A] The total molecular mass of the entire substituent group.
- [B] The atomic number of the atoms directly attached to the double-bond carbons (higher atomic number = higher priority).
- [C] The alphabetical order of substituent names.
- [D] The number of hydrogen atoms in the substituent.

FREQUENTLY EXAMINED CORE QUESTIONS — HIGH-FREQUENCY REPEATS (Q101 – Q110)

101 [HARD] [HIGH-FREQUENCY EXAM REPEAT] HF 1: Determining Empirical and Molecular Formulas from Percentage Composition — 9701/12/M/J/22/Q13 [1]

An organic liquid X has the percentage composition by mass: C, 62.07%; H, 10.34%; O, 27.59%. In a mass spectrometer, the molecular ion peak M^+ occurs at $m/z = 116.0$.

What is the molecular formula of compound X?

- [A] C_3H_6O
- [B] $C_5H_8O_3$
- [C] $C_6H_{12}O_2$
- [D] $C_4H_8O_2$

102 [HARD] [HIGH-FREQUENCY EXAM REPEAT] HF 2: Identifying the Number of Chiral Centers in Complex Drug Molecules — 9701/11/O/N/21/Q15 [1]

Menthol is a cyclic organic compound with the structural formula shown below:
2-isopropyl-5-methylcyclohexan-1-ol

How many chiral carbon atoms (stereocenters) are present in one molecule of menthol?

- [A] 3 chiral centers (C1 bearing -OH, C2 bearing isopropyl, and C5 bearing methyl)
- [B] 1 chiral center
- [C] 2 chiral centers
- [D] 4 chiral centers

103 [HARD] [HIGH-FREQUENCY EXAM REPEAT] HF 3: Systematic IUPAC Naming with Multiple Functional Groups — 9701/13/M/J/22/Q14 [1]

What is the systematic IUPAC name for $CH_3-CH(OH)-CH_2-CH=CH_2$?

- [A] Pent-1-en-4-ol
- [B] 4-hydroxypent-1-ene
- [C] 2-hydroxypent-4-ene
- [D] Pent-4-en-2-ol

104 [HARD] [HIGH-FREQUENCY EXAM REPEAT] HF 4: Total Number of Isomers with Formula C_4H_8 — 9701/12/F/M/22/Q16 [1]

Including both structural isomers and stereoisomers (cis-trans), how many isomeric compounds have the molecular formula C_4H_8 ?

- [A] 4
- [B] 6 (but-1-ene, cis-but-2-ene, trans-but-2-ene, 2-methylpropene, cyclobutane, and methylcyclopropane)
- [C] 5
- [D] 3

105 [HARD] [HIGH-FREQUENCY EXAM REPEAT] HF 5: Hybridisation of Carbon Atoms Across an Unsaturated Chain — 9701/11/M/J/21/Q14 [1]

Consider the compound $CH_2=CH-C\equiv C-CH_3$.

From left to right, what are the orbital hybridisations of the five carbon atoms?

- [A] sp^2 , sp^2 , sp , sp , sp^3
- [B] sp^2 , sp^2 , sp^2 , sp^2 , sp^3
- [C] sp^3 , sp^2 , sp , sp , sp^3
- [D] sp , sp , sp^2 , sp^2 , sp^3

106 [HARD] [HIGH-FREQUENCY EXAM REPEAT] HF 6: Mechanism of Free Radical Substitution — Identification of Termination Products — 9701/12/O/N/17/Q15

During the reaction between methane and chlorine in the presence of ultraviolet light, trace amounts of ethane, C₂H₆, are detected in the product mixture.

Which reaction accounts for the formation of ethane?

- [A] CH₄ + .CH₃ → C₂H₆ + .H
- [B] .CH₃ + Cl₂ → C₂H₆ + 2Cl.
- [C] .CH₃ + .CH₃ → C₂H₆ (a termination step between two methyl radicals)
- [D] CH₃Cl + CH₃Cl → C₂H₆ + Cl₂

107 [HARD] [HIGH-FREQUENCY EXAM REPEAT] HF 7: Distinguishing Enantiomers Using Physical and Chemical Properties — 9701/13/O/N/21/Q16 [1]

Which property is DIFFERENT for two optical enantiomers of lactic acid (2-hydroxypropanoic acid)?

- [A] Their boiling points at standard atmospheric pressure.
- [B] The direction in which they rotate the plane of plane-polarized light in a polarimeter.
- [C] Their solubility in pure water at 25 °C.
- [D] Their acid dissociation constant (K_a) in aqueous solution.

108 [HARD] [HIGH-FREQUENCY EXAM REPEAT] HF 8: Calculating Sigma and Pi Bonds in Complex Molecules — 9701/11/F/M/23/Q15 [1]

Acrylonitrile (propenenitrile), CH₂=CH-C#N, is used in the manufacture of acrylic fibers.

How many sigma (sigma) bonds and pi (pi) bonds are present in one molecule of acrylonitrile?

- [A] 5 sigma bonds and 4 pi bonds
- [B] 7 sigma bonds and 2 pi bonds
- [C] 4 sigma bonds and 5 pi bonds
- [D] 6 sigma bonds and 3 pi bonds

109 [HARD] [HIGH-FREQUENCY EXAM REPEAT] HF 9: Predicting Stereoisomerism in Dienes — 9701/12/M/J/23/Q16 [1]

How many geometric (cis-trans) stereoisomers exist for hexa-2,4-diene, CH₃-CH=CH-CH=CH-CH₃?

- [A] 3 stereoisomers (cis-cis, trans-trans, and cis-trans)
- [B] 4 stereoisomers
- [C] 2 stereoisomers
- [D] 1 stereoisomer

110 [HARD] [HIGH-FREQUENCY EXAM REPEAT] HF 10: Deductive Identification of Structural Isomers with C₃H₈O — 9701/13/M/J/23/Q17 [1]

Three isomeric organic liquids, P, Q, and R, all have the molecular formula C₃H₈O.

- P and Q react with sodium metal to evolve hydrogen gas.
- P can be oxidized by acidified potassium dichromate(VI) to an aldehyde.
- R does NOT react with sodium metal.

What are compounds P, Q, and R?

- [A] P is propan-2-ol, Q is propan-1-ol, and R is propanoic acid
- [B] P is propan-1-ol, Q is methoxyethane, and R is propan-2-ol
- [C] P is propan-1-ol, Q is propan-2-ol, and R is methoxyethane
- [D] P is methoxyethane, Q is propan-1-ol, and R is propan-2-ol

QUICK-CHECK ANSWER KEY GRID (Q1 – Q110)

Official Cambridge 9701 Paper 1 Marking Matrix · 1 Mark Per Question · Total Marks: 110

Q1 – Q20		Q21 – Q40		Q41 – Q60		Q61 – Q80		Q81 – Q100	
Q01	B	Q21	B	Q41	B	Q61	B	Q81	B
Q02	D	Q22	D	Q42	D	Q62	D	Q82	D
Q03	A	Q23	A	Q43	A	Q63	A	Q83	A
Q04	C	Q24	C	Q44	C	Q64	C	Q84	C
Q05	A	Q25	A	Q45	A	Q65	A	Q85	A
Q06	D	Q26	D	Q46	D	Q66	D	Q86	D
Q07	B	Q27	B	Q47	B	Q67	B	Q87	B
Q08	C	Q28	C	Q48	C	Q68	C	Q88	C
Q09	B	Q29	B	Q49	B	Q69	B	Q89	B
Q10	A	Q30	A	Q50	A	Q70	A	Q90	A
Q11	D	Q31	D	Q51	D	Q71	D	Q91	D
Q12	C	Q32	C	Q52	C	Q72	C	Q92	C
Q13	A	Q33	A	Q53	A	Q73	A	Q93	A
Q14	C	Q34	C	Q54	C	Q74	C	Q94	C
Q15	B	Q35	B	Q55	B	Q75	B	Q95	B
Q16	D	Q36	D	Q56	D	Q76	D	Q96	D
Q17	C	Q37	C	Q57	C	Q77	C	Q97	C
Q18	A	Q38	A	Q58	A	Q78	A	Q98	A
Q19	D	Q39	D	Q59	D	Q79	D	Q99	D
Q20	B	Q40	B	Q60	B	Q80	B	Q100	B

FREQUENTLY EXAMINED CORE QUESTIONS (Q101 – Q110)

Q101 C	Q102 A	Q103 D	Q104 B	Q105 A	Q106 C	Q107 B	Q108 D	Q109 A	Q110 C
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COMPREHENSIVE EXAMINER EXPLANATIONS & DISTRACTOR ANALYSIS

Q	KEY	EXAMINER SCIENTIFIC RATIONALE & KEY CONCEPT TESTED	MARKS
1	B	Option B is correct. Moles of C = $54.5 / 12.0 = 4.54$; moles of H = $9.1 / 1.0 = 9.10$; moles of O = $36.4 / 16.0 = 2.275$. Dividing by smallest (2.275): C: $4.54/2.275 = 2.0$; H: $9.10/2.275 = 4.0$; O: $2.275/2.275 = 1.0$. Thus, empirical formula is C_2H_4O (empirical formula mass = $2(12) + 4(1) + 16 = 44.0$). Since $M_r = 88.0$, the molecular formula multiplier is $88.0 / 44.0 = 2$, giving $C_4H_8O_2$.	[1]
2	D	Option D is correct. In skeletal notation, carbon-carbon bonds are drawn as lines in a zig-zag format. Carbon atoms and attached hydrogen atoms are not explicitly written: each vertex (corner) and each line termination represents a carbon atom together with enough implicit hydrogen atoms to give that carbon a total of four covalent bonds.	[1]
3	A	Option A is correct. Number the longest continuous carbon chain (5 carbons = pentane) from the end that gives substituents the lowest locant set. Numbering from left: C2 has methyl, C3 has bromo (locants 2,3). Numbering from right: C3 has bromo, C4 has methyl (locants 3,4). Lower locant set is (2,3). Substituents are cited alphabetically: 'bromo' before 'methyl'. Hence: 3-bromo-2-methylpentane.	[1]
4	C	Option C is correct. Both aliphatic aldehydes and aliphatic ketones have the general formula $C_nH_{2n}O$ with one degree of unsaturation (a carbonyl group $C=O$). Because they contain no carbon-carbon double bonds ($C=C$), they do not decolorize aqueous bromine by addition.	[1]
5	A	Option A is correct. $HO-C_6H_4-$ is a phenolic hydroxyl group attached directly to an aromatic benzene ring; $-CH(OH)-$ is attached to two carbon atoms (the ring and CH_2), making it a secondary alcohol; $-NH-$ is bonded to two alkyl carbons ($-CH_2-$ and $-CH_3$), classifying it as a secondary amine.	[1]
6	D	Option D is correct. All members of a homologous series possess the same functional group and general formula, differing by successive $-CH_2-$ units. This produces similar chemical properties and regular gradations in physical properties like boiling point.	[1]
7	B	Option B is correct. A displayed formula shows every single atom and every covalent bond (single, double, or triple) in the molecule explicitly.	[1]
8	C	Option C is correct. The longest carbon chain with the $C=C$ bond has 4 carbons (but-2-ene). Numbering from the left gives the methyl substituent the lowest locant: 2-methylbut-2-ene.	[1]
9	B	Option B is correct. An ester contains a carbonyl group bonded directly to an alkoxy oxygen: $-C(=O)O-$.	[1]
10	A	Option A is correct. Ratio: C = $85.7/12 = 7.14$; H = $14.3/1 = 14.3$. Ratio 1:2 $\rightarrow$ empirical CH_2 (mass 14). $M_r = 56 / 14 = 4 \rightarrow C_4H_8$.	[1]
11	D	Option D is correct. A 4-carbon unbranched chain with carboxylic acid groups at both ends is butanedioic acid.	[1]
12	C	Option C is correct. 2,2-dimethylpropane (neopentane) has a central carbon bonded to four methyl groups: $C(CH_3)_4$.	[1]
13	A	Option A is correct. Nitriles contain a carbon-nitrogen triple bond: $-C\equiv N$.	[1]
14	C	Option C is correct. 4-carbon chain with chloro substituents at carbons 2 and 3: 2,3-dichlorobutane.	[1]
15	B	Option B is correct. Formula: $C + 1 - (H/2) = 6 + 1 - 5 = 2$. This corresponds to two double bonds, one triple bond, or combinations of rings and double bonds.	[1]
16	D	Option D is correct. Acyl chlorides contain the $-C(=O)Cl$ functional group, as in CH_3COCl (ethanoyl chloride).	[1]
17	C	Option C is correct. Longest chain containing $-OH$ is 4 carbons (butan-2-ol). Methyl branch at C2: 2-methylbutan-2-ol.	[1]
18	A	Option A is correct. In 2-methylpropan-2-ol, the carbon bearing the $-OH$ group is bonded to three other carbon atoms (three methyl groups), making it tertiary.	[1]
19	D	Option D is correct. Amino acids contain both a carboxylic acid group ($-COOH$) and an amine group ($-NH_2$) attached to the α -carbon.	[1]
20	B	Option B is correct. Open-chain alkanes possess the general formula C_nH_{2n+2} .	[1]
21	B	Option B is correct. Ring formation eliminates two hydrogens compared to open-chain alkanes, giving C_nH_{2n} .	[1]
22	D	Option D is correct. A saturated six-carbon ring with single $C-C$ bonds is cyclohexane (C_6H_{12}).	[1]
23	A	Option A is correct. CH_3COCH_3 (propanone) contains a carbonyl group bonded to two alkyl carbons, characteristic of a ketone.	[1]
24	C	Option C is correct. An oxygen atom bonded between two alkyl groups ($R-O-R'$) is an ether (methoxyethane).	[1]
25	A	Option A is correct. The carbonyl carbon of the aldehyde is always numbered C1. The chain is 4 carbons (butanal) with methyl at C3: 3-methylbutanal.	[1]
26	D	Option D is correct. Benzene (C_6H_6) is depicted as a hexagon with a circle representing the delocalized 6 π electrons.	[1]
27	B	Option B is correct. Moles of C = $0.80 / 0.20 = 4$; moles of H = $(0.80 \times 2) / 0.20 = 8$. Hydrocarbon is C_4H_8 .	[1]

Q	KEY	EXAMINER SCIENTIFIC RATIONALE & KEY CONCEPT TESTED	MARKS
28	C	Option C is correct. The 3-carbon carboxylate ion is propanoate, so the salt is sodium propanoate.	[1]
29	B	Option B is correct. The systematic IUPAC name for a 3-carbon chain with an amino group at carbon-1 is propan-1-amine.	[1]
30	A	Option A is correct. Propenoic acid contains both a carbon-carbon double bond (alkene) and a carboxylic acid group.	[1]
31	D	Option D is correct. In homolytic fission (homolysis), the covalent bond breaks evenly. Each bonding partner leaves with one electron from the shared pair (depicted by single-headed curly arrows / fish-hooks), resulting in the generation of two highly reactive, neutral species with unpaired electrons known as free radicals: $X-Y \rightarrow X\cdot + Y\cdot$.	[1]
32	C	Option C is correct. An electrophile ('electron-lover') is an electron-poor species (such as Br^+ , NO_2^+ , H^+ , or the δ^+ carbon in a carbonyl group) that seeks out electron-rich centers (like the pi bond of an alkene or benzene ring) and accepts a lone pair of electrons to form a new covalent bond.	[1]
33	A	Option A is correct. A nucleophile ('nucleus-lover') is an electron-rich species containing at least one non-bonding lone pair of electrons (e.g., hydroxide $:OH^-$, cyanide $:CN^-$, ammonia $:NH_3$, water $:OH_2$). It attacks electron-deficient carbon centers (δ^+) and donates its lone pair to form a new dative/covalent bond.	[1]
34	C	Option C is correct. When CH_3CH_2Br is heated with ethanolic KOH (where OH^- acts as a Bronsted-Lowry base rather than a nucleophile), a hydrogen halide molecule (HBr) is removed from adjacent carbon atoms to form a C=C double bond ($CH_2=CH_2 + KBr + H_2O$). This removal of atoms without replacement is an elimination reaction.	[1]
35	B	Option B is correct. A propagation step consumes one free radical and generates a new free radical, maintaining the radical chain: (1) $Cl\cdot + CH_4 \rightarrow \cdot CH_3 + HCl$; (2) $\cdot CH_3 + Cl_2 \rightarrow CH_3Cl + Cl\cdot$. Option A is initiation (0 radicals $\rightarrow$ 2 radicals). Options C and D are termination steps (2 radicals $\rightarrow$ 0 radicals).	[1]
36	D	Option D is correct. A full double-barbed curly arrow depicts the movement of an electron pair from an electron-rich site (lone pair or bond) to an electron-deficient site.	[1]
37	C	Option C is correct. A single-headed (fish-hook) arrow indicates the movement of a single electron, frequently employed in free radical mechanisms.	[1]
38	A	Option A is correct. Alkyl groups are electron-donating via inductive effect (+I), dispersing the positive charge on the carbon. Tertiary carbocations have three alkyl groups providing maximum charge dispersal, making $3^\circ > 2^\circ > 1^\circ > \text{methyl}$.	[1]
39	D	Option D is correct. Alkenes possess a pi bond with high electron density that attacks electrophiles like H-Br, undergoing electrophilic addition.	[1]
40	B	Option B is correct. Haloalkanes contain a polar C-Br bond attacked by nucleophilic $:OH^-$ ions, substituting the halogen atom.	[1]
41	B	Option B is correct. Hydrolysis is a bond-breaking chemical reaction involving water, e.g., the hydrolysis of esters into carboxylic acids and alcohols.	[1]
42	D	Option D is correct. In organic synthesis, condensation involves the joining of two molecules accompanied by the loss of a small molecule like water or hydrogen chloride.	[1]
43	A	Option A is correct. $NaBH_4$ acts as a source of hydride ions ($:H^-$), reducing aldehydes to primary alcohols and ketones to secondary alcohols.	[1]
44	C	Option C is correct. Organic oxidation involves the gain of oxygen bonds (e.g., alcohol $\rightarrow$ aldehyde $\rightarrow$ carboxylic acid) or the loss of hydrogen bonds.	[1]
45	A	Option A is correct. Alkyl groups push electron density along sigma bonds toward electron-deficient centers (+I inductive effect), stabilizing carbocations.	[1]
46	D	Option D is correct. The electrophilic addition of HX to an unsymmetrical alkene generates two possible carbocations; the proton adds to the carbon with more hydrogens to form the more stable (more alkyl-substituted) carbocation.	[1]
47	B	Option B is correct. SN_1 (Substitution Nucleophilic Unimolecular) proceeds via rate-determining loss of leaving group to form a carbocation intermediate, followed by rapid attack of nucleophile. Favored by 3° substrates.	[1]
48	C	Option C is correct. SN_2 proceeds via simultaneous nucleophile attack from the rear and leaving group departure, causing inversion of configuration (Walden inversion).	[1]
49	B	Option B is correct. A vertical condenser cools escaping vapors back to liquid, allowing continuous heating at boiling point without solvent loss.	[1]
50	A	Option A is correct. The photon energy of UV light matches the bond dissociation enthalpy of Cl-Cl or Br-Br, causing homolytic fission to start the chain reaction.	[1]
51	D	Option D is correct. Addition combines all reactants into one product (pi bond breaks, two sigma bonds form); substitution replaces one atom/group with another.	[1]
52	C	Option C is correct. Termination occurs when any two free radicals collide and combine their unpaired electrons to form a covalent bond, extinguishing the radicals.	[1]
53	A	Option A is correct. Oxygen's higher electronegativity draws pi and sigma electrons toward itself ($C(\delta^+) = O(\delta^-)$), leaving the carbonyl carbon electron-deficient and vulnerable to nucleophilic attack.	[1]

Q	KEY	EXAMINER SCIENTIFIC RATIONALE & KEY CONCEPT TESTED	MARKS
54	C	Option C is correct. Backside attack in SN2 requires space around the carbon; in 3° haloalkanes, the three bulky alkyl groups cause extreme steric hindrance, blocking nucleophile access.	[1]
55	B	Option B is correct. High temperature, ethanolic solvent, and high base concentration favor elimination (forming propene) over substitution (forming propan-2-ol).	[1]
56	D	Option D is correct. Each carbon in ethene forms three sigma bonds (two C-H and one C-C) by mixing one 2s and two 2p orbitals to form three sp ² hybrid orbitals in a planar trigonal arrangement with 120° angles. The unhybridised 2p orbital on each carbon overlaps sideways to form the pi bond.	[1]
57	C	Option C is correct. Sigma bonds result from direct end-on overlap along the line connecting the nuclei, giving maximum overlap and cylindrical electron density. Pi bonds form by weaker lateral overlap of parallel p orbitals, concentrating electron density in two lobes above and below the internuclear plane. This sideways overlap prevents free rotation.	[1]
58	A	Option A is correct. Each carbon in H-C#C-H forms two sp hybrid orbitals at 180° (one sigma to H, one sigma to C). Total sigma bonds = 2 (C-H) + 1 (C-C) = 3 sigma bonds. The two unhybridised p orbitals on each carbon overlap sideways to produce 2 perpendicular pi bonds. Geometry is linear (180°).	[1]
59	D	Option D is correct. Carbons 1 and 2 are sp ² hybridised with trigonal planar geometry (120°). Carbon 3 is sp ³ hybridised with tetrahedral geometry (109.5°).	[1]
60	B	Option B is correct. Propanone has 6 C-H sigma bonds, 2 C-C sigma bonds, 1 C-O sigma bond, and 1 C=O pi bond. Total = 9 sigma bonds and 1 pi bond.	[1]
61	B	Option B is correct. The carbonyl carbon forms three sigma bonds using sp ² hybrid orbitals oriented at ~120° in a plane, with the pi bond formed by unhybridised p orbital overlap.	[1]
62	D	Option D is correct. A pi bond requires parallel alignment of the overlapping p orbitals. Twisting around the C-C axis breaks this sideways overlap, requiring significant energy (~260 kJ/mol), which cannot occur at room temperature.	[1]
63	A	Option A is correct. Each of the 6 carbon atoms in benzene is sp ² hybridised and contributes one electron from its unhybridised 2p orbital, forming a continuous delocalised pi cloud of 6 electrons.	[1]
64	C	Option C is correct. As the number of shared electron pairs increases, electrostatic attraction for the two nuclei increases, drawing them closer together: C-C (0.154 nm) > C=C (0.134 nm) > C#C (0.120 nm).	[1]
65	A	Option A is correct. Triple bond involves sharing 6 electrons (one sigma and two pi bonds) with bond energy ~837 kJ/mol, double bond is ~612 kJ/mol, and single bond is ~348 kJ/mol.	[1]
66	D	Option D is correct. In the -C#N group, both carbon and nitrogen form one sigma bond and hold two pi bonds (or one lone pair on N), corresponding to sp hybridisation.	[1]
67	B	Option B is correct. Ethene is completely planar; all 6 atoms (2 carbons and 4 hydrogens) lie in the same spatial plane due to sp ² hybridisation.	[1]
68	C	Option C is correct. In diamond, each carbon is sp ³ hybridised forming 4 tetrahedral sigma bonds. In graphite, each carbon is sp ² hybridised forming 3 planar trigonal sigma bonds with delocalised pi electrons.	[1]
69	B	Option B is correct. Formula C ₄ H ₆ : 6 C-H sigma bonds + 3 C-C sigma bonds = 9 sigma bonds (plus 2 pi bonds).	[1]
70	A	Option A is correct. Oxygen has four electron pairs (two bonding pairs and two non-bonding lone pairs) in a tetrahedral arrangement. Lone pair repulsion reduces the C-O-H angle to ~104.5° - 108.5°.	[1]
71	D	Option D is correct. Nitrogen has three single bonding pairs and one non-bonding lone pair. The electron pair geometry is tetrahedral, yielding a trigonal pyramidal molecular geometry (~107°).	[1]
72	C	Option C is correct. Each double bond consists of 1 sigma and 1 pi bond. Thus, O=C=O contains 2 sigma bonds and 2 pi bonds.	[1]
73	A	Option A is correct. The +CH ₃ carbocation carbon has three bonding electron pairs and an empty 2p orbital, resulting in sp ² hybridisation and planar geometry with 120° bond angles.	[1]
74	C	Option C is correct. Due to pi electron delocalization, all six C-C bonds in benzene are identical in length (0.139 nm), intermediate between pure single and pure double bonds.	[1]
75	B	Option B is correct. The pi bond in the C=O group arises from the lateral (sideways) overlap of parallel 2p orbitals on carbon and oxygen.	[1]
76	D	Option D is correct. Structural isomers have the same molecular formula but different structural formulas (different connectivity of atoms). The three categories are: (1) Chain isomerism (different carbon skeleton branching); (2) Positional isomerism (same skeleton, functional group in different locations); (3) Functional group isomerism (same formula, different functional group).	[1]
77	C	Option C is correct. Both propanal (CH ₃ CH ₂ CHO) and propanone (CH ₃ COCH ₃) have molecular formula C ₃ H ₆ O, but propanal contains an aldehyde functional group (-CHO) whereas propanone contains a ketone group (>C=O). They are functional group isomers.	[1]
78	A	Option A is correct. Geometric (cis-trans) isomerism requires: (1) Restricted rotation about the C=C bond (due to the pi bond); and (2) Each individual carbon atom of the C=C bond must be bonded to two non-identical groups (i.e. Cab=Ccd, where a! = b and c! = d). If either carbon has two identical groups (e.g. Ca ₂ =Ccd), cis-trans isomerism is impossible.	[1]

Q	KEY	EXAMINER SCIENTIFIC RATIONALE & KEY CONCEPT TESTED	MARKS
79	D	Option D is correct. A chiral center (asymmetric carbon) is bonded tetrahedrally to four distinct atoms or groups. This asymmetry means the molecule lacks a plane of symmetry, creating two non-superimposable mirror image structures (enantiomers) that rotate the plane of plane-polarized light in opposite directions.	[1]
80	B	Option B is correct. In butan-2-ol, C2 is bonded to four different groups: -H, -OH, -CH ₃ , and -CH ₂ CH ₃ . Thus, C2 is a chiral stereocenter, and butan-2-ol exists as two enantiomers. In all other isomers, every carbon has at least two identical groups (-H ₂ , -H ₃ , or -(CH ₃) ₂).	[1]
81	B	Option B is correct. C ₄ H ₁₀ has exactly two structural chain isomers: unbranched butane (CH ₃ CH ₂ CH ₂ CH ₃) and branched 2-methylpropane ((CH ₃) ₃ CH).	[1]
82	D	Option D is correct. C ₅ H ₁₂ has three chain isomers: pentane, 2-methylbutane (isopentane), and 2,2-dimethylpropane (neopentane).	[1]
83	A	Option A is correct. The chlorine atoms can be 1,2 (ortho), 1,3 (meta), or 1,4 (para), yielding three positional isomers.	[1]
84	C	Option C is correct. Both aliphatic carboxylic acids (R-COOH) and esters (R-COOR') share the general formula C _n H _{2n} O ₂ and are functional group isomers.	[1]
85	A	Option A is correct. Cis-trans isomerism requires two different groups on EACH carbon of the C=C bond. In but-1-ene, C1 is bonded to two identical hydrogen atoms (=CH ₂), so swapping them produces the same molecule.	[1]
86	D	Option D is correct. In cis-1,2-dichloroethene, both polar C-Cl bonds point to the same side, creating a net molecular dipole and higher boiling point (60 °C). In trans, the dipoles point in opposite directions and cancel completely (net dipole = 0), lowering the boiling point (48 °C).	[1]
87	B	Option B is correct. In 2-bromobutane, C2 is bonded to -H, -Br, -CH ₃ , and -CH ₂ CH ₃ (four different groups), making it chiral.	[1]
88	C	Option C is correct. A racemate contains equal amounts of (+) and (-) enantiomers. The optical rotation caused by one isomer is precisely cancelled by the equal and opposite rotation of the other, resulting in optical inactivity.	[1]
89	B	Option B is correct. A molecule with n chiral centers and no internal symmetry has 2 ⁿ stereoisomers. For n=1, 2 ¹ = 2 enantiomers.	[1]
90	A	Option A is correct. Enantiomers have identical physical properties (boiling point, density, refractive index, solubility in achiral solvents) except that they rotate the plane of plane-polarized light in equal and opposite directions (+ / d- vs - / l-).	[1]
91	D	Option D is correct. Pent-2-ene has a C=C bond where C2 is bonded to -H and -CH ₃ , and C3 is bonded to -H and -CH ₂ CH ₃ . Since both carbons have two different groups, it exists as cis-pent-2-ene and trans-pent-2-ene.	[1]
92	C	Option C is correct. Carbon-3 is bonded to four different groups: -H, -CH ₃ (methyl), -CH ₂ CH ₃ (ethyl), and -CH ₂ CH ₂ CH ₃ (propyl).	[1]
93	A	Option A is correct. Both molecules possess the same 3-carbon chain and the same alcohol functional group (-OH), but the group is at C1 in propan-1-ol and C2 in propan-2-ol (positional isomers).	[1]
94	C	Option C is correct. Pentane (straight 5-carbon chain) and 2,2-dimethylpropane (branched chain) have the same formula C ₅ H ₁₂ but different carbon skeleton branching, making them chain isomers.	[1]
95	B	Option B is correct. The four isomers are: 1,1-dichloropropane, 1,2-dichloropropane, 1,3-dichloropropane, and 2,2-dichloropropane.	[1]
96	D	Option D is correct. Open-chain alkenes (with C=C functional group) and ring cycloalkanes (saturated cyclic rings) share formula C _n H _{2n} and are functional group isomers.	[1]
97	C	Option C is correct. The rigid three-membered ring restricts rotation, enabling cis and trans isomers. Furthermore, the trans isomer lacks an internal plane of symmetry and exists as a pair of chiral enantiomers.	[1]
98	A	Option A is correct. In glycine, the central alpha-carbon is attached to -NH ₂ , -COOH, and TWO -H atoms. Because two groups are identical, it is achiral and optically inactive.	[1]
99	D	Option D is correct. C ₂ H ₆ O has two isomers: ethanol (an alcohol) and methoxymethane / dimethyl ether (an ether). They belong to different functional group classes, exhibiting functional group isomerism.	[1]
100	B	Option B is correct. The CIP priority rules rank substituents based on the atomic number of the atom directly bonded to the stereocenter (e.g. -Br (35) > -Cl (17) > -OH (8) > -CH ₃ (6) > -H (1)). If high-priority groups are on opposite sides, it is E (entgegen); on the same side, it is Z (zusammen).	[1]
101	C	Option C is correct. Moles of C = 62.07/12.0 = 5.1725; moles of H = 10.34/1.0 = 10.34; moles of O = 27.59/16.0 = 1.7244. Divide by smallest (1.7244): C: 5.1725/1.7244 = 3.0; H: 10.34/1.7244 = 6.0; O: 1.7244/1.7244 = 1.0. Empirical formula = C ₃ H ₆ O (formula mass = 3(12) + 6(1) + 16 = 58.0). Since M _r = 116.0, multiplier = 116.0 / 58.0 = 2. Molecular formula = C ₆ H ₁₂ O ₂ .	[1]
102	A	Option A is correct. Analyzing the cyclohexane ring: C1 is bonded to -H, -OH, C2, and C6 (different pathways around the ring); C2 is bonded to -H, -CH(CH ₃) ₂ , C1, and C3; C5 is bonded to -H, -CH ₃ , C4, and C6. Each of these three carbons has four distinctly different groups attached. Thus, menthol contains exactly 3 chiral centers.	[1]
103	D	Option D is correct. Under IUPAC nomenclature rules, the principal functional group suffix (alcohol, -ol) takes precedence over the alkene unsaturation (-en-) for determining the lowest locant. Numbering from the right would give the alcohol locant 4, while numbering from the left gives the alcohol locant 2 (pent-4-en-2-ol). Hence, the correct systematic name is pent-4-en-2-ol.	[1]

Q	KEY	EXAMINER SCIENTIFIC RATIONALE & KEY CONCEPT TESTED	MARKS
104	B	Option B is correct. Formula C_4H_8 has 1 degree of unsaturation. Alkenes: (1) but-1-ene, (2) cis-but-2-ene, (3) trans-but-2-ene, (4) 2-methylpropene. Cycloalkanes: (5) cyclobutane, (6) methylcyclopropane. Total number of isomers = 6.	[1]
105	A	Option A is correct. C1 ($CH_2=$) forms 3 sigma bonds $\rightarrow$ sp^2 ; C2 ($=CH-$) forms 3 sigma bonds $\rightarrow$ sp^2 ; C3 ($-C\equiv$) forms 2 sigma bonds $\rightarrow$ sp ; C4 ($\equiv C-$) forms 2 sigma bonds $\rightarrow$ sp ; C5 ($-CH_3$) forms 4 sigma bonds $\rightarrow$ sp^3 . Sequence: sp^2, sp^2, sp, sp, sp^3 .	[1]
106	C	Option C is correct. In the propagation steps, a high concentration of methyl free radicals ($\cdot CH_3$) is generated. When two methyl radicals collide, their single unpaired electrons pair up to form a C-C covalent bond ($\cdot CH_3 + \cdot CH_3 \rightarrow C_2H_6$). This radical-radical collision consumes both radicals without generating new ones, representing a classic termination step.	[1]
107	B	Option B is correct. Enantiomers possess identical scalar physical properties (boiling point, melting point, density, K_a , solubility in achiral solvents) because the internal interatomic distances and energies are mirror-symmetric. They differ ONLY in their interaction with chiral phenomena: the direction in which they rotate plane-polarized light (+ vs -) and their reaction rates with other chiral reagents/enzymes.	[1]
108	D	Option D is correct. Counting bonds: 3 C-H single bonds = 3 sigma; 1 C=C double bond = 1 sigma + 1 pi; 1 C-C single bond = 1 sigma; 1 C#N triple bond = 1 sigma + 2 pi. Total sigma bonds = $3 + 1 + 1 + 1 = 6$ sigma bonds. Total pi bonds = $1 + 2 = 3$ pi bonds.	[1]
109	A	Option A is correct. Hexa-2,4-diene has two C=C bonds, each capable of cis (Z) or trans (E) configurations. Due to the symmetrical nature of the carbon chain, the (cis, trans) and (trans, cis) arrangements are identical by 180° rotation. Thus, there are only 3 distinct geometric isomers: (1) cis,cis-hexa-2,4-diene; (2) trans,trans-hexa-2,4-diene; (3) cis,trans-hexa-2,4-diene.	[1]
110	C	Option C is correct. C_3H_8O has 3 structural isomers: propan-1-ol, propan-2-ol, and methoxyethane ($CH_3-O-C_2H_5$). Alcohols react with sodium metal to form alkoxides and H_2 gas ($2ROH + 2Na \rightarrow 2RONa + H_2$), so P and Q are alcohols. Primary alcohols oxidize to aldehydes, identifying P as propan-1-ol (1° alcohol) and Q as propan-2-ol (2° alcohol). Ethers lack O-H bonds and do not react with sodium, confirming R as methoxyethane.	[1]